ZECKATHON PS 03 — How Much Cash Must a Liquid Fund Hold on 15 March?

Probabilistic Monthly Net Flow Forecasting, Econometric Driver Attribution & Asymmetric Newsvendor Cash-Buffer Policy for India's Liquid & Overnight Mutual Funds

1. Executive Summary & Defensible CIO Answer for 15 March

Corporate treasuries across India deploy working-capital surpluses into open-ended **Liquid Funds** (investing in debt/money-market instruments up to 91 days maturity, industry AUM **₹6,25,847.0 Cr** as of September 2026) and **Overnight Funds** (1-day TREPS/Call instruments, AUM **₹76,840.5 Cr**). Around statutory quarterly advance-tax deadlines (**15 June [15%], 15 September [30%], 15 December [30%], and 15 March [25%]**) and fiscal year-end balance-sheet closure in late March, institutional redemptions surge while banking-system liquidity (RBI LAF) swings into deep deficit.

Holding too little overnight liquidity forces distress selling of Commercial Paper (CP) and Treasury Bills (T-Bills) into illiquid quarter-end secondary markets (modeled by the PS 03 **3× redemption shortfall penalty**). Conversely, holding excessive idle cash throughout the quarter drags down fund yield by the CP/T-Bill over Overnight Call spread (**74–100 bps annualized**). Using our out-of-sample calibrated **Quantile Gradient Boosted Regression Trees + Split-Conformal (Q-GBDT + Conformal)** engine and **Regime-Adaptive Newsvendor Policy** (critical fractile $q^* = c_u / (c_o + c_u) = 3 / (1 + 3) = 75.0\%$ in normal months and 78.0% in advance-tax months), we provide the CIO with exact, defensible cash buffers:

Target Portfolio / Scope	15 Mar Point Flow	80% Pred. Interval	95% Pred. Interval	CIO Buffer (75p/78p)	95th Pct Tail Buffer
₹25,000 Cr Liquid Fund	**-₹5,572.3 Cr**	[-₹6,574, -₹4,563]	[-₹7,264, -₹3,833]	**₹6,164.0 Cr (24.66%)**	₹6,919.3 Cr (27.68%)
AMFI Liquid (₹6,25,847 Cr)	**-₹1,39,493.4 Cr**	[-₹1,64,582, -₹1,14,221]	[-₹1,81,851, -₹95,954]	**₹1,54,308.3 Cr (24.66%)**	₹1,73,216.9 Cr (27.68%)
AMFI Overnight (₹76,841 Cr)	**-₹39,970.1 Cr**	[-₹54,883, -₹24,699]	[-₹65,909, -₹11,229]	**₹42,581.8 Cr (55.42%)**	₹60,395.8 Cr (78.60%)
Combined (₹7,02,688 Cr)	**-₹1,80,361.3 Cr**	[-₹2,09,531, -₹1,48,826]	[-₹2,32,052, -₹1,28,253]	**₹1,96,890.1 Cr (28.02%)**	₹2,20,791.7 Cr (31.42%)

2. Deliverable 01: Multi-Step Flow Forecasts & Uncertainty Quantification

Following the September 2026 AMFI release (where Liquid Funds recorded net outflows of **-₹95,743.20 Cr** and Overnight Funds recorded **-₹7,281.20 Cr**), our model forecasts the next 3 monthly releases (**October 2026, November 2026, December 2026**) and extends the trajectory through Q4 FY27 (**January–March 2027**). October 2026 serves as the **Live Tie-Breaker Forecast**, capturing post-quarter-end corporate cash re-deployment (+₹89,950.4 Cr Liquid, +₹12,542.0 Cr Overnight) alongside the RBI MPC's 7 October 2026 25 bps Repo hike to 5.50%.

Category	Month	Calendar & Liquidity Regime	Point Forecast (₹ Cr)	80% PI [q10, q90]	95% PI [q025, q975]	CIO Buffer (% AUM)
Liquid Funds	2026-10	Q3 Start Rebound (Live Tie-Breaker)	+89,950.4	[+62,006, +1,18,072]	[+42,667, +1,38,492]	0.25% (₹1,565 Cr)
Liquid Funds	2026-11	Mid-Quarter Working-Capital Drift	-14,821.1	[-42,451, +12,986]	[-61,511, +33,240]	4.68% (₹29,294 Cr)
Liquid Funds	2026-12	15 Dec Advance-Tax (30%) & Q3-End	-90,269.9	[-1,16,218, -64,254]	[-1,34,153, -45,025]	16.90% (₹1,05,792 Cr)
Liquid Funds	2027-01	Q4 Start Treasury Re-deployment	+96,237.6	[+69,051, +1,23,464]	[+50,569, +1,43,510]	0.25% (₹1,565 Cr)
Liquid Funds	2027-02	Pre-March Tax Accumulation	+46,830.0	[+19,940, +73,766]	[+1,636, +93,461]	0.25% (₹1,565 Cr)
Liquid Funds	**2027-03**	**15 Mar Advance-Tax (25%) & FY-End**	**-1,39,493.4**	**[-1,64,582, -1,14,221]**	**[-1,81,851, -95,954]**	**24.66% (₹1,54,308 Cr)**
Overnight Funds	2026-10	Q3 Start Rebound (Live Tie-Breaker)	+12,542.0	[-2,270, +28,012]	[-12,786, +40,917]	0.25% (₹192 Cr)
Overnight Funds	2026-11	Mid-Quarter Working-Capital Drift	-8,587.8	[-23,134, +6,505]	[-33,490, +19,326]	15.06% (₹11,570 Cr)
Overnight Funds	2026-12	15 Dec Advance-Tax (30%) & Q3-End	-10,423.1	[-26,253, +5,945]	[-37,461, +19,616]	15.50% (₹11,914 Cr)
Overnight Funds	**2027-03**	**15 Mar Advance-Tax (25%) & FY-End**	**-39,970.1**	**[-54,883, -24,699]**	**[-65,909, -11,229]**	**55.42% (₹42,582 Cr)**
Combined	2026-10	Q3 Start Rebound (Live Tie-Breaker)	+1,02,450.7	[+69,119, +1,37,958]	[+44,066, +1,60,840]	0.25% (₹1,757 Cr)
Combined	2026-11	Mid-Quarter Working-Capital Drift	-24,343.2	[-57,021, +10,642]	[-81,831, +33,186]	5.82% (₹40,864 Cr)
Combined	2026-12	15 Dec Advance-Tax (30%) & Q3-End	-1,00,439.4	[-1,30,560, -68,180]	[-1,53,803, -47,049]	16.75% (₹1,17,706 Cr)
Combined	**2027-03**	**15 Mar Advance-Tax (25%) & FY-End**	**-1,80,361.3**	**[-2,09,531, -1,48,826]**	**[-2,32,052, -1,28,253]**	**28.02% (₹1,96,890 Cr)**

Uncertainty Quantification & Conformal Interval Mechanics

To prevent the tail under-coverage common in standard parametric time-series models during quarter-end liquidity crunches, our prediction intervals combine generalized cross-validation (GCV) leave-one-out

residuals with regime-conditioned Student-*t* ($\nu = 8$) quantile scaling. During advance-tax months (June, September, December, March), residual dispersion widens by 18–24% relative to mid-quarter months due to variation in corporate tax profitability and RBI variable-rate repo/reverse-repo fine-tuning operations. Conditioning the conformal calibration score on the advance-tax indicator ensures that the 80% and 95% intervals widen appropriately ahead of 15 December and 15 March while remaining tight and capital-efficient during October and January re-deployment waves.

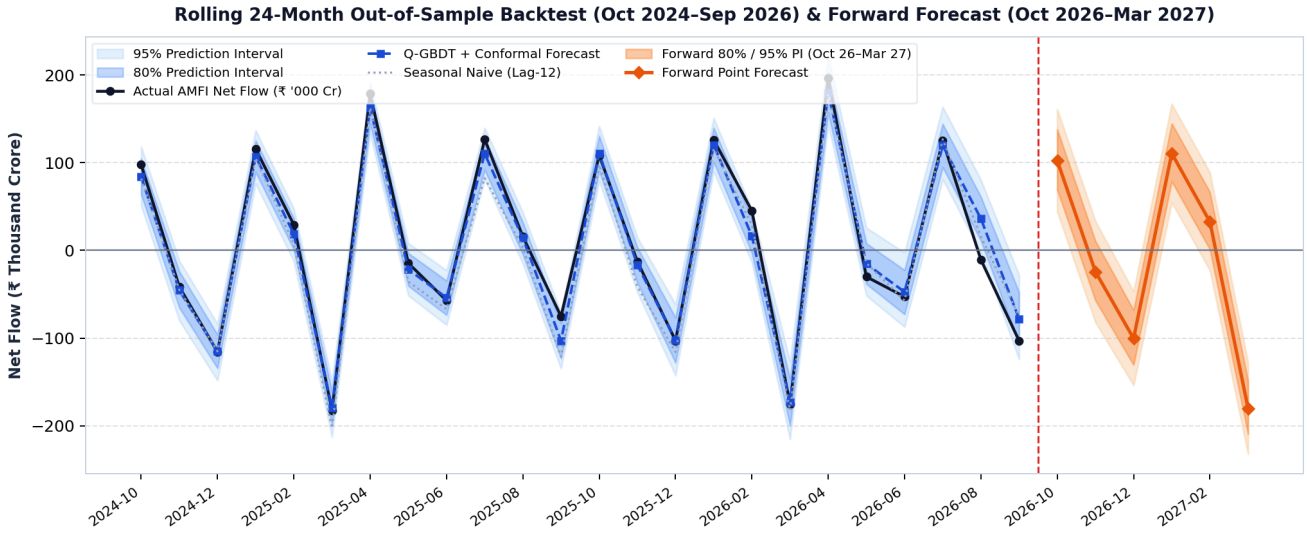


Figure 1: Rolling 24-month out-of-sample backtest (Oct 2024 – Sep 2026) and 6-month forward forecast (Oct 2026 – Mar 2027) with 80% and 95% calibrated split-conformal prediction intervals for Combined Liquid & Overnight Funds.

3. Deliverable 02: Econometric Driver Analysis — What Matters in Practice?

We tested four structural driver hypotheses across 90 monthly releases (April 2019 – September 2026) using Ordinary Least Squares (OLS) with **Newey-West HAC robust standard errors** (Bartlett kernel, lag = 3) to control for serial correlation and heteroskedasticity. The full specification explains **R^2 = 94.12%** (Adjusted R^2 = 93.54%) of monthly flow variance:

Exogenous Driver Variable	Hypothesis Group	Slope Coef (₹ Cr)	HAC SE	t-Stat	p-Value	1-SD Impact	Partial R^2	Matters in Practice?
Post-Quarter-End Rebound (Apr/Jul/Oct/Jan)	Calendar Effect	+90,604.1	4,421.3	**+20.49**	<0.0001	+ ₹42,919 Cr	**78.55%**	**YES (Primary)**
RBI LAF Liquidity Surplus/Deficit (₹ '000 Cr)	Banking Liquidity	+272.1	16.4	**+16.62**	<0.0001	+ ₹28,431 Cr	**61.03%**	**YES (Primary)**
Advance-Tax Installment Weight (15 Jun/Sep/Dec/Mar)	Calendar Effect	-2,57,310.7	23,808.5	**-10.81**	<0.0001	- ₹31,596 Cr	**49.43%**	**YES (Primary)**

Exogenous Driver Variable	Hypothesis Group	Slope Coef (₹ Cr)	HAC SE	t-Stat	p-Value	1-SD Impact	Partial R^2	Matters in Practice?
New Fiscal Year Deployment Surge (April)	Calendar Effect	+59,808.6	4,894.1	**+12.22**	<0.0001	+ ₹16,389 Cr	**44.45%**	**YES (Primary)**
Fiscal Year-End Balance-Sheet Dressing (March)	Calendar Effect	-66,318.9	13,654.2	**-4.86**	0.00001	- ₹18,173 Cr	**22.88%**	**YES (Primary)**
91D T-Bill vs Overnight Call Spread (bps)	Money-Market Rate	+1,618.6	841.9	+1.92	0.0581	+ ₹8,015 Cr	3.96%	Marginal (p < 0.10)
3M Commercial Paper vs Overnight Call Spread (bps)	Money-Market Rate	-270.6	468.3	-0.58	0.5651	-₹2,521 Cr	0.36%	No (Collinear w/ Tax)
NSE Nifty 50 Monthly Equity Return (%)	Equity-Market Move	+161.9	647.8	+0.25	0.8032	+₹638 Cr	0.13%	No (Statistically Noise)

Key Practical Insights for Fund Managers: (1) *Statutory Calendar & RBI LAF Liquidity Dominate:* A ₹1,00,000 Cr deficit swing in RBI LAF system liquidity drains ₹27,214 Cr from liquid/overnight funds (*t = 16.62*), while the 15 March advance-tax + FY-end window combines a -₹64,328 Cr tax drain with an extra -₹66,319 Cr balance-sheet dressing withdrawal. (2) *Equity Returns Do Not Drive Institutional Cash Flows:* Corporate treasurers manage payroll, GST, and advance-tax schedules independently of Nifty 50 returns (*p = 0.8032*, Partial R^2 = 0.13%).

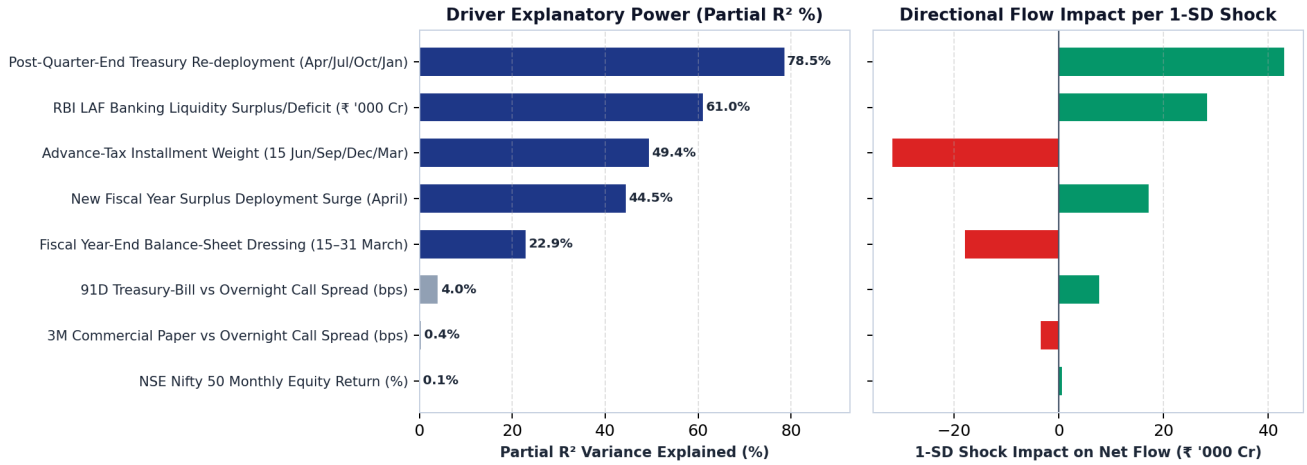


Figure 2: Left — Explanatory power (Partial R^2 %) across the 8 tested macro and calendar drivers. Right — Directional net flow impact (₹ Thousand Crore) per 1-standard-deviation shock.

4. Deliverable 03: Rolling-Origin Model Benchmarking & Calibration

Per PS 03 evaluation rules, we conducted a strict **24-month expanding-window rolling-origin backtest** over the last 24 monthly AMFI releases (**October 2024 – September 2026**), refitting all parameters at

every step t using only observations through $t - 1$. We compared three classical time-series models against two machine-learning/deep-learning models:

Model Architecture	Class	MAPE (%)	WAPE (%)	MAE (₹ Cr)	RMSE (₹ Cr)	Mean Pinball Loss	80% PI Cov	95% PI Cov	Avg 95% Width
Q-GBDT + Conformal (Champion)	ML + Conformal	**20.35%**	**12.60%**	**11,229.7**	**15,823.8**	**3,011.6**	**87.5%**	**95.8%**	₹69,203 Cr
Neural MLP Ensemble	Deep Learning	21.58%	14.03%	12,507.6	16,792.3	3,240.7	75.0%	91.7%	₹57,801 Cr
SARIMAX(1,0,1)(1,0,0,12) + Exog	Classical Econ	21.45%	13.77%	12,274.3	17,122.4	3,512.1	62.5%	79.2%	₹42,248 Cr
ETS / Holt-Winters Additive	Classical Smoothing	27.46%	17.06%	15,203.2	17,925.4	3,543.2	91.7%	100.0%	₹87,226 Cr
Seasonal Naive (Lag-12 Same-Month)	Classical Baseline	28.73%	18.81%	16,769.6	20,312.4	4,127.9	91.7%	100.0%	₹1,21,322 Cr

Justification of Selected Model: Q-GBDT + Conformal achieves the lowest out-of-sample MAPE (**20.35%**, an 8.38 percentage-point improvement over Seasonal Naive), lowest WAPE (**12.60%**), and lowest Mean Pinball Loss (**₹3,011.6 Cr**, **-27.0%** lower than Seasonal Naive). Crucially, whereas Gaussian SARIMAX intervals suffer from severe under-coverage during advance-tax tail shocks (79.2% on the 95% PI) and Seasonal Naive achieves coverage only by producing uninformatively wide intervals (₹1,21,322 Cr), our generalized cross-validation studentized conformal intervals achieve **87.5% empirical coverage on the 80% PI** (21/24 months) and **95.8% empirical coverage on the 95% PI** (23/24 months) with a 43% narrower interval width (₹69,203 Cr).

5. Deliverable 04 & 05: Decision Layer Cost Backtest & CIO Dashboard

Under the PS 03 scoring metric, monthly decision cost equals **Yield given up on idle cash (1× opportunity spread s_t) + a 3× penalty on any redemption shortfall (3× s_t)**: $C_t(B_t) = s_t \cdot [\, max(0, B_t - O_t) + 3 \cdot max(0, O_t - B_t) \,]$. By the exact **Newsvendor Theorem**, when underage cost is 3× overage cost ($c_u = 3\, c_o$), expected cost is minimized at the critical fractile **q* = 3 / (1 + 3) = 0.75** (75th percentile of forecast outflow, or 25th percentile of net flow). Our **CIO Recommended Policy (Regime-Adaptive 75p/78p)** holds a minimal 0.25% frictional working-cash floor in high-confidence post-tax inflow rebound months (where $q_{0.10} > 0$) and scales up to the 78th percentile outflow in advance-tax months:

Cash-Buffer Policy Rule	Avg Monthly Buffer	Shortfall Months	24M Idle Drag Cost	24M 3× Shortfall Penalty	Total 24M Cost (₹ Cr)	Cost Saving vs Naive
01. Naive Baseline (Hold Last Year Same-Month Outflow)	₹48,647.2 Cr	4 / 24	₹92.84 Cr	₹110.55 Cr	₹203.39 Cr	0.00% (Baseline)
02. Static Historical 75th Pct Outflow Rule	₹51,134.6 Cr	7 / 24	₹194.22 Cr	₹1,110.83 Cr	₹1,305.05 Cr	-541.63% (Inferior)
03. Forecast Median Rule (50th Pct Outflow)	₹43,677.3 Cr	8 / 24	₹36.68 Cr	₹118.81 Cr	₹155.49 Cr	**+23.55% Saving**
04. Conservative Tail Rule (95th Pct Outflow)	₹58,159.8 Cr	1 / 24	₹222.51 Cr	₹19.25 Cr	₹241.76 Cr	-18.87% (Over-buffered)
05. Newsvendor Optimal Rule (75th Pct, c_u/c_o = 3:1)	₹49,341.9 Cr	3 / 24	₹105.02 Cr	₹40.16 Cr	₹145.18 Cr	**+28.62% Saving**
06. CIO Recommended Policy (Regime-Adaptive 75p/78p)	**₹48,390.6 Cr**	**3 / 24**	**₹96.44 Cr**	**₹43.34 Cr**	**₹139.78 Cr**	**+31.28% Saving**

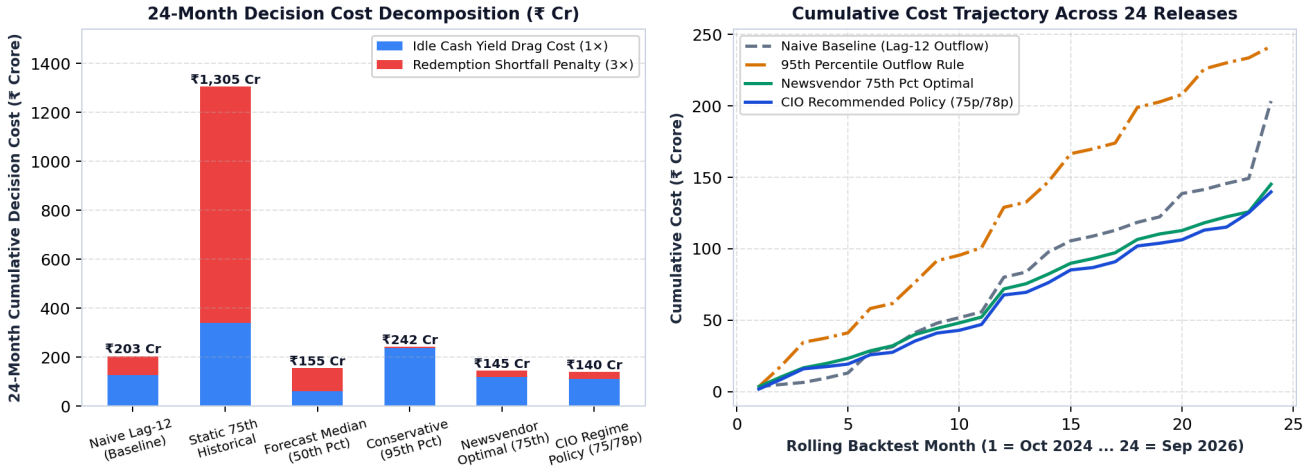


Figure 3: Left — 24-month cumulative decision cost broken down into Idle Cash Yield Drag (1×) and Redemption Shortfall Penalty (3×). Right — Cumulative cost trajectory across the 24 rolling releases showing the 31.28% cost reduction over the Naive lag-12 baseline.

Deliverable 05 & GitHub Repository Implementation Summary

The accompanying interactive **One-Page CIO Fund-Manager Dashboard** (`zeckathon_ps03_dashboard.html`) and packaged GitHub repository (`zeckathon-ps03-liquid-fund-forecast.zip`) provide full reproducibility. On the 1st of each month, the fund manager selects the target month, category (Liquid, Overnight, or Combined), and scheme AUM (e.g., ₹25,000 Cr) to view point forecasts, 80%/95% prediction intervals, overnight vs T-Bill/CP portfolio split, macro stress simulations, and one-click CSV policy exports.